Local Spectral Similarity-Guided Sparse Unmixing of Hyperspectral Images With Spatial Graph Regularization

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***Abstract*— As the spectral library continues to expand, sparse hyperspectral unmixing methods have been developed to solve the mixing problem without the need for end-member extraction or generation. These methods leverage the intrinsic spectral and spatial information to enhance the accuracy of fractional abundance estimation. However, their effectiveness is limited by rigid spatial regularization and insufficient utilization of spectral–spatial information that hampers the improvement of unmixing performance. To overcome this limitation, we present a novel algorithm named local spectral similarity-guided sparse hyperspectral unmixing with spatial graph regularization (SGSU). In SGSU, we introduce a spatial graph regularization to enforce the interpixel correlation within spatial clusters and assign them to corresponding abundance vectors. To reduce the computational cost, we employ an adaptive superpixel-based spatial grouping strategy to segment the hyperspectral image, which translates the intrinsic geometry into constraints on abundance. Furthermore, we introduce a weighting factor with two components into the sparse unmixing framework. One component is based on the row sparsity of the estimated abundances, indicating the presence of active end-members; the other component is based on the similarity between neighboring pixels, which promotes piecewise smoothness of the estimated abundances. To obtain a more robust solution, we adopt a double-loop scheme based on the alternating direction method of multipliers (ADMM) algorithm to solve the SGSU model. Experimental results on both simulated and real hyperspectral datasets demonstrate that the proposed SGSU algorithm outperforms state-of-the-art sparse unmixing methods in terms of both accuracy of abundance estimation and end-member identification from spectral libraries. Our algorithm achieves superior unmixing results, which indicates its potential for practical applications in hyperspectral imaging.**

***Index Terms*— Graph regularization, sparse hyperspectral unmixing, spatial information, spatially weighted unmixing, superpixel segmentation.**

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I. Introduction

HYPERSPECTRAL imaging technology captures images of the Earth's surface through hundreds of wavelength channels, making it possible to detect distinct spectral features. This technique has been widely used in various fields, such as mineral exploration, environmental monitoring, precision agriculture, target detection, and medical applications [1]. Nevertheless, the distribution of terrestrial objects can be complex and diverse and the spatial resolution of imaging spectrometers is not always high enough to capture fine details of these objects. This can result in the presence of mixed pixels in hyperspectral remote-sensing images, which makes it difficult to accurately analyze and interpret the data. Hyperspectral unmixing estimates the end-members present in each mixed pixel and the fractional abundance of each end-member, which enables the identification and quantification of materials in a scene.

As public spectral libraries continue to grow, mixed pixels can be represented as a combination of multiple entries in the known library [2], [3], [4]. Since mixed pixels involve few end-members compared to the spectral library entries, the corresponding abundance vectors are sparse, which results in sparse hyperspectral unmixing [5], [6], [7]. The goal of this technique is to find the optimal subset of spectral features (from a large spectral library) that each mixed pixel in the hyperspectral image can best represent. As a semisupervised method, sparse unmixing overcomes the drawback of virtual end-members [8] introduced by unsupervised unmixing and avoids estimating the number of end-members.

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A sparse unmixing algorithm with variable splitting and extended Lagrangian (SUnSAL) was proposed. This method implemented sparsity in the abundance solution for the first time [5]. A collaborative SUnSAL (CLSUnSAL) was then proposed to further constrain the row sparsity of the abundance matrix to reduce the number of end-members forming the hyperspectral image [9]. Furthermore, introducing weighting factors into the regularizer is effective in penalizing nonzero coefficients in the abundance solution and improving the sparsity of the solution [10]. For example, the iterative reweighted sparse unmixing (IRWSU) [11] and double reweighted sparse unmixing (DRSU) [12] algorithms were developed to promote more sparsity of the solution with promising results. However, these methods do not take into account the spatial information in the image, and exploiting spatial (prior) information has been recognized as an important way to improve unmixing results [13].

Considering that the pixels in the image are spatially correlated, a sparse unmixing method using spatial regularization has been proposed and the obtained abundances are more consistent. Sparse unmixing by variable splitting augmented Lagrangian and total variation (SUnSAL-TV) is one of the best-known unmixing methods from [14]. Total variation (TV) spatial regularization smooths adjacent pixels and promotes spatial uniformity of images via first-order neighborhood systems. Based on TV regularization, the DRSU-TV algorithm improves both the sparsity of fractional abundances and the spatial smoothness of abundance maps [15]. In addition, spatial weighting factors are also used to characterize spatial information by introducing weights into the sparse regulator. In [16], local collaborative sparse unmixing (LCSU) uses spatial weighting to obtain spatial structure information in local areas. In [17], the spectral–spatial weighted sparse unmixing (S^2WSU) framework introduces two weighting factors containing both spectral and spatial information to make better use of neighborhood information. These spatially constrained sparse unmixing algorithms show great potential for improving unmixing performance.

Additionally, note that similar pixels within a local region may have similar end-members and abundances. This property of hyperspectral images has led to the development of graph-based approaches for hyperspectral unmixing. Graph regularization maps hyperspectral data onto a weighted graph. Here, a node represents a pixel spectrum and a weighted edge between two nodes represents the similarity of two pixel spectra. Graph topology can promote similar data smoothness in arbitrary neighborhood structures, while weight values preserve detail in the reconstructed image. For example, the sparsity-constrained graph ℓ_1-Laplacian-regularized NMF algorithm (SC-GLRℓ_1NMF) uses graph structures to embed spatial and spectral relationships in the nonnegative matrix factorization (NMF) framework. It facilitates partial smoothing and reduces noise [18]. The double reweighted sparse graph-regularized hyperspectral unmixing algorithm (DRSGHU) adds a global graph regularization to the reweighted ℓ_1 sparse unmixing framework. It utilizes the rich structural information of the entire hyperspectral image and imposes sparsity constraints on abundance solutions [19]. Superpixel-based graph Laplacian regularization (SBGLSU) builds a graph Laplacian matrix with presegmented homogeneous regions and subsequently adopts block-wise graph regularization to obtain spatial similarity [20]. These methods exploit the potentially diverse structure of hyperspectral images and map it onto abundance maps by constructing graph regulators, showing great potential to promote higher-order relationships between adjacent pixels [21].

To fully exploit the inherent spectral–spatial information and the underlying structure of hyperspectral images, we develop a novel spatial graph regularization (SGSU) sparse hyperspectral unmixing algorithm based on local spectral similarity. As is well known, pixels within a local region are generally composed of similar materials in similar proportions. The proposed SGSU algorithm takes advantage of the sparsity of the abundance matrix and the diverse structure of the original hyperspectral data by introducing weighting factors and flexibly applies Laplace regularization to the graph. In our approach, on the one hand, the structure of the hyperspectral image is mapped to estimated abundances by a graph regularizer. The graph regularizer groups highly correlated neighboring pixels into a graph to model the underlying higher-order structure. On the other hand, spatial weighting measures the spectral similarity of different pixels within a neighborhood, while spectral weighting emphasizes the concentration of nonzero abundance values in several matrix rows and the sparseness of tiny abundance values in the image. Moreover, the proposed SGSU model is efficiently solved using a two-loop iterative scheme based on the alternating direction method of multipliers (ADMM) [22]. The main contributions of the SGSU algorithm introduced in this article can be summarized as follows.

1) A spatial weighting factor relying on the similarity of neighborhood pixels is introduced into the ℓ_1 sparse unmixing framework to facilitate piecewise smoothness of estimated abundances. The SGSU algorithm, proposed as an extension of the SBGLSU algorithm [20], incorporates spatial weights to enhance the spatial consistency among adjacent pixels. Furthermore, it differs from the spatial weight utilized in the S^2WSU algorithm [17] in that it measures the spatial neighborhood information through local spectral similarity, while the S^2WSU algorithm quantifies the importance of neighboring pixels based on Euclidean distance. Experimental results indicate that the introduced SGSU algorithm, even with only the spatial weighted sparse regularization, significantly enhances the accuracy of abundance estimation.
2) The newly proposed SGSU algorithm incorporates a superpixel-based spatial graph regularization, which aims to maintain the internal manifold structure of the hyperspectral image and constrain the correlation of pixels within spatial clusters. Moreover, by integrating dual-weighting factors for both spatial and spectral information, it effectively incorporates local spatial and spectral characteristics into its regularization framework. This enhances its ability to accurately restore the abundance of information on the original substances

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in various hyperspectral image scenes, effectively coping with diverse scene conditions, and improving the generalizability of unmixing.

The rest of the article is organized as follows. Section II provides a brief introduction to the linear mixing model (LMM) and the sparse unmixing algorithm. The proposed SGSU unmixing method (and its corresponding optimization solution algorithm) is presented in Section III. Sections IV and V describe the results of experiments performed on simulated and real hyperspectral datasets and provide a comprehensive performance evaluation. Finally, Section VI summarizes this work and provides directions for future research.

II. Relevant Background Works

This section provides a brief introduction to the LMM and discusses related sparse unmixing models.

A. Linear Mixing Model

An LMM assumes that mixed pixels in a hyperspectral image can be viewed as a set of pure spectral signatures (end-members)

$$\mathbf{y} = \mathbf{M}\mathbf{x} + \mathbf{n} \tag{1}$$

where $\mathbf{y} = [y_1, y_2, \ldots, y_L]^T \in \mathbb{R}^{L\times 1}$ represents a pixel vector of the observed hyperspectral data, L is the number of bands, $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_q] \in \mathbb{R}^{L\times q}$ represents the end-member matrix, q is the number of spectral signatures, $\mathbf{x} = [x_1, x_2, \ldots, x_q]^T \in \mathbb{R}^{q\times 1}$ represents the abundance vector with respect to end-member matrix $\mathbf{M}$, and $\mathbf{n} = [n_1, n_2, \ldots, n_L]^T \in \mathbb{R}^{L\times 1}$ is an error vector and additive noise. Generally, the fractional abundance needs to satisfy two physical constraints: the abundance nonnegativity constraint (ANC) (i.e., $x_i \geq 0, i = 1, 2, \ldots, q$), and the abundance sum-to-one constraint (ASC) (i.e., $\sum_{i=1}^{q} x_i = 1$), respectively.

As spectral libraries are constantly growing, the linear sparse unmixing model avoids the shortcomings of end-member identification by introducing the end-member spectral library $\mathbf{A}$ as prior knowledge. In other words, the traditional LMM end-member matrix $\mathbf{M}$ is replaced by a redundant spectral dictionary $\mathbf{A}$. The model can be rewritten as

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n} \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{L\times m}$ is a spectral dictionary with m pure spectral signatures.

B. Sparse Unmixing Model

Note that the spectral library has many more end-members than mixed pixels actually have, so there are many zeros in the corresponding weighting factors. In other words, the abundance matrix associated with the redundant library is inherently sparse. Therefore, the $\ell_2-\ell_0$ framework is naturally suited to the sparse unmixing problem of finding the optimal set of end-members from a spectral library. The model can be expressed as

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{A}\mathbf{X} - \mathbf{Y}||_F^2 + \lambda||\mathbf{X}||_0, \quad \text{s.t. } \mathbf{X} \geq 0 \tag{3}$$

where $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_n] \in \mathbb{R}^{L\times n}$ represents the observed data with n pixel vectors, $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in \mathbb{R}^{m\times n}$ represents the abundance matrix, $\|\cdot\|_F$ represents the Frobenius norm, $\lambda \geq 0$ is a regularization parameter, and $\mathbf{X} \geq 0$ indicates compliance with ANC. Note that ANC is explicitly enforced while ASC is not, as some comments have pointed out that ASC may have a negative impact [5].

However, the objective function (3) is a nonconvex problem that is difficult to solve directly [23], [24]. SUnSAL uses the $\ell_2 - \ell_1$ norm instead of the $\ell_2 - \ell_0$ norm, and the sparse separation problem translates to [22]

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{A}\mathbf{X} - \mathbf{Y}||_F^2 + \lambda||\mathbf{X}||_{1,1}, \quad \text{s.t. } \mathbf{X} \geq 0 \tag{4}$$

where $||\mathbf{X}||_{1,1} = \sum_{j=1}^{n} ||\mathbf{x}_j||_1$ and $\mathbf{x}_j$ represents the jth column of $\mathbf{X}$. The optimization problem related to (4) has been solved by SUnSAL with the ADMM [22]. However, the simple ℓ_1 regularizer indiscriminately constrains each abundance vector in $\mathbf{X}$, ignoring the spatial correlation between pixels, making it difficult to achieve the desired level of sparsity.

C. Double Reweighted Sparse Unmixing

To better characterize the sparsity of the solution, the DRSU model [12] introduces a double weight to select fewer end-members from the dictionary and enhances the sparsity of the abundance vector associated with each pixel. The model is expressed as

$$\min_{\mathbf{X}} \frac{1}{2}||\mathbf{A}\mathbf{X} - \mathbf{Y}||_F^2 + \lambda||(\mathbf{W}_2\mathbf{W}_1) \odot \mathbf{X}||_{1,1}, \quad \text{s.t. } \mathbf{X} \geq 0 \tag{5}$$

where $\odot$ is an operator that computes the element-wise multiplication of two variables, $\mathbf{W}_1 \in \mathbb{R}^{m\times n}$ and $\mathbf{W}_2 \in \mathbb{R}^{m\times m}$ are dual-weights, $\mathbf{W}_1$ penalizes small nonzero values and enlarges large fractional abundances [10], $\mathbf{W}_2$ acts on the rows in the abundance matrix so that nonzero values are clustered in a few rows, $||(\mathbf{W}_2\mathbf{W}_1) \odot \mathbf{X}||_{1,1}$ is a regularizer that introduces a sparsity prior on abundances, and λ is the regularization parameter. In [12], the optimization of Problem (5) adopts an iterative scheme based on ADMM. At the $t+1$th iteration, the two weights $\mathbf{W}_1$ and $\mathbf{W}_2$ are calculated as follows:

$$\mathbf{W}_{1,ij}^{(t+1)} = \frac{1}{\mathbf{X}_{ij}^{(t)} + \varepsilon} \tag{6}$$

where $\varepsilon > 0$ is a small constant

$$\mathbf{W}_2^{(t+1)} = \text{diag}\left[\frac{1}{||\mathbf{X}^{(t)}(1,:)||_2 + \varepsilon}, \ldots, \frac{1}{||\mathbf{X}^{(t)}(m,:)||_2 + \varepsilon}\right] \tag{7}$$

where $\mathbf{X}^{(t)}(i,:)$ is the ith row vector in $\mathbf{X}$ at iteration t. From (6) and (7), it can be derived that small abundance coefficients will generate large weights, such that small values in the coefficient tend to zero in the next iteration, and the sparsity of the solution is enhanced.

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III. Proposed Approach

In this section, we present the proposed SGSU model and first show its relationship to existing models. Then, we solve the related optimization problem based on the ADMM.

A. Local Spectral Similarity-Guided Sparse Hyperspectral Unmixing Model With Spatial Graph Regularization

A sparse representation simply characterizes the abundance sparsity for a spectral library. However, it does not address the underlying geometry of hyperspectral images. It is well known that pixels in both spatial and spectral neighborhoods are likely to have the same or similar distribution of endmembers, thus yielding similar abundances. One can take advantage of this property to limit abundance estimation and improve the accuracy of the sparse unmixing process. To do this, we consider both the correlation of adjacent pixels in spatial space and the similarity of pixels in spectral space. In particular, a hyperspectral image is divided into multiple superpixels based on spectral and spatial information, such that the set of abundance vectors corresponding to each superpixel retains the diverse structure of the original set of spectral signatures. Furthermore, the spectral similarity of pixels within local square blocks is introduced into the sparse regularization, leading to an estimate of the central pixel abundance. These two regularizations bring a similar relationship of spatially and spectrally adjacent pixels into the separation problem. Given a hyperspectral image $\mathbf{Y} \in \mathbb{R}^{L\times n}$ with L bands of n pixels. Consider each pixel $\mathbf{y}$ in the image $\mathbf{Y}$ as a node and establish an edge between any two nodes to create a graph. Each edge of the graph indicates the spectral similarity between two pixels. There are several methods to define the weight matrix $\mathbf{W} \in \mathbb{R}^{n\times n}$ of a neighborhood graph, among which binary weights, heat kernel weights, and dot product weights are three commonly used approaches [25]. Considering the characteristics of hyperspectral data in this context, the heat kernel weighting method is more suitable for defining the weight matrix $\mathbf{W} \in \mathbb{R}^{n\times n}$

$$\mathbf{W}_{ij} = e^{-\frac{\|y_i - y_j\|^2}{2\sigma^2}} \tag{8}$$

where $\mathbf{W}_{ij}$ denotes the weight of the edge between pixels $\mathbf{y}_i$ and $\mathbf{y}_j$, and σ denotes the bandwidth of the kernel. It can be seen from (8) that if the pixels $\mathbf{y}_i$ and $\mathbf{y}_j$ are closer, that is, their spectral similarity is higher, the value of the weight $\mathbf{W}_{ij}$ will be larger, tending to be 1, while the smaller $\mathbf{W}_{ij}$ means that $\mathbf{y}_i$ and $\mathbf{y}_j$ are less similar.

The weighted graph reflects the underlying structure of the hyperspectral image, expected to be transformed into abundance space. However, pixels within local blocks tend to share similarities, resulting in similar abundances. Note that the graph regularization term in traditional methods requires computing the global adjacency matrix, which imposes a huge computational burden for large-size image processing, posing great challenges for practical applications. To address this issue, we use the simple linear iterative clustering (SLIC) [26] method to divide the image into multiple small regions with good continuity and high internal similarity. As a variant of the K-means clustering method, SLIC only searches for similar pixels within a limited region, and both the spectral distance and the Euclidean distance between pixels serve as measures of similarity. The definition of distance is given in [27]. When constructing the adjacency matrix, only the within-region pixel similarity is considered, which can significantly reduce the computational burden and facilitate the practical application of the algorithm. A superpixel-based graph Laplacian regularizer is designed as follows:

$$\begin{aligned}&\frac{1}{2}\sum_{i,j=1}^{n_p} \|\mathbf{x}_i^{(p)} - \mathbf{x}_j^{(p)}\|_2^2 \mathbf{W}_{pij} \\ &\quad= \sum_{i=1}^{n_p} \mathbf{x}_i^{(p)^T} \mathbf{x}_i^{(p)} \mathbf{D}_{pii} - \sum_{i,j=1}^{n_p} \mathbf{x}_i^{(p)^T} \mathbf{x}_j^{(p)} \mathbf{W}_{pij} \\ &\quad= \mathbf{Tr}\left(\mathbf{X}_p \mathbf{D}_p \mathbf{X}_p^T\right) - \mathbf{Tr}\left(\mathbf{X}_p \mathbf{W}_p \mathbf{X}_p^T\right) = \mathbf{Tr}\left(\mathbf{X}_p \mathbf{L}_p \mathbf{X}_p^T\right)\end{aligned} \tag{9}$$

where n_p denotes the number of pixels in the pth superpixel, $\mathbf{x}_i^{(p)}$ and $\mathbf{x}_j^{(p)}$ denote any abundance vector in the pth superpixel, $\mathbf{X}_p$ denotes the abundance submatrix associated with the pth superpixel, $\mathbf{W}_p$ denotes the adjacency matrix of pth superpixel, $\mathbf{W}_{pij}$ denotes each entry of $\mathbf{W}_p$, $\mathbf{L}_p = \mathbf{D}_p - \mathbf{W}_p$ is the graph Laplacian matrix of the pth superpixel, $\mathbf{D}_{pii} = \sum_{j=1}^{p} \mathbf{W}_{pij}$, and $\mathrm{Tr}(\cdot)$ denotes the trace of a matrix.

To simultaneously exploit the structural information from the hyperspectral data and the sparsity of the abundance matrix, we developed a local spectral similarity-guided sparse hyperspectral unmixing model using spatial graph regularization (SGSU), as follows:

$$\begin{aligned}\min_{\mathbf{X}} \ &\frac{1}{2}||\mathbf{AX} - \mathbf{Y}||_F^2 + \lambda||(\mathbf{H}_1\mathbf{H}_2) \odot \mathbf{X}||_{1,1} \\ &+ \tau \sum_{s=1}^{p} \mathbf{Tr}\left(\mathbf{X}_s \mathbf{L}_s \mathbf{X}_s^T\right), \quad \text{s.t. } \mathbf{X} \geq 0\end{aligned} \tag{10}$$

where p is the number of superpixels in the hyperspectral image. $\mathbf{X}_s$ is the abundance submatrix corresponding to the sth superpixel. $\lambda \geq 0$ and $\tau \geq 0$ are the regularization parameters. The first term is the data fit term measured by the Frobenius norm, the second term is the sparsity regularization term aimed at promoting the sparsity of the abundance matrix, and the third term explores the hidden information in the hyperspectral data, where the interabundance is integrated to examine the correlation of the vector. Inspired by the favorable behavior of the weighted ℓ_1 norm, we develop two weighting factors to more efficiently introduce spatial and spectral information into the model for the purpose of sparse unmixing. Let $\mathbf{H}_1 \in \mathbb{R}^{m\times m}$ be the spectral weighting coefficients updated in iteration $t+1$ as follows:

$$\mathbf{H}_1^{(t+1)} = \mathrm{diag}\left[\frac{1}{||\mathbf{X}^{(t)}(1,:)||_2 + \varepsilon}, \ldots, \frac{1}{||\mathbf{X}^{(t)}(m,:)||_2 + \varepsilon}\right]. \tag{11}$$

As for the weight matrix $\mathbf{H}_1$, it builds on the success of [9] and [12] and the common sparsity of all abundance vectors makes the objective that it is carried out collaboratively. In fact, it promotes row sparsity of estimated abundances. Given the importance of integrating spatial information for unmixing, we usually assume that there is a high spatial correlation

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between adjacent pixels in hyperspectral images, implying piecewise smoothing of abundance maps. Considering this information, similar to [17]'s S^{2}WSU algorithm, the local spectral similarity-guided spatial weighting factor $\mathbf{H}_2 \in \mathbb{R}^{m\times n}$. Repeat $h_{2,ij}^{(t+1)}$ at line i of $\mathbf{H}_2$ at iteration $t+1$. If the element is located in column j, the spatial weighting factor is expressed as

$$h_{2,ij}^{(t+1)} = \frac{1}{f_{b\in\mathcal{N}(j)}\left(x_{ib}^{(t)}\right)+\varepsilon} \tag{12}$$

where $\mathcal{N}(j)$ is the adjacency set of elements x_{ij}, x_{ib} is any element of $\mathcal{N}(j)$, and $f(\cdot)$ is a function that measures the spectral similarity between an element x_{ij} and its neighbors, as well as spatial correlations by neighboring systems. In spatial neighborhoods, considering that pixels whose spectra are close together also have more similar abundance vectors, $f(\cdot)$ is expressed as

$$f(x_{ij}) = \frac{\sum_{b\in\mathcal{N}(j)}\theta_{ib}x_{ib}}{\sum_{b\in\mathcal{N}(j)}\theta_{ib}} \tag{13}$$

where $\theta_{ib} = (1/||\mathbf{x}_b - \mathbf{x}_j||_2)$ is a spectral similarity metric between $\mathbf{x}_j$ and its neighborhood abundance vector $\mathbf{x}_b$, $\mathbf{x}_j$ is the central vector, and x_b is an abundance vector composed of elements $\mathbf{x}_{ib}$. θ_{ib} represents neighbor importance, where eight-connectivity (3×3 windows) is used for the following algorithm design and applications. It should be noted that formulas (12) and (13) were inspired by the equations in [17], but they exhibit distinctions between each other. Especially for formula (12), the SGSU algorithm incorporates local spectral similarity to evaluate spatial neighborhood information for the spatial weighting factor, whereas the S^{2}WSU algorithm quantifies the significance of neighboring pixels based on Euclidean distance.

B. Optimization Algorithm

The weighted ℓ_1 problem can be optimized by ADMM [12], [15], but at the same time, the weight computation makes it difficult to prove ADMM convergence in this scenario. To this end, we implement an iterative updating scheme using inner and outer double-loops to solve the optimization problem. As shown in (10), the inner loop uses ADMM to solve a set of variables and the outer loop computes two weights $\mathbf{H}_1$ and $\mathbf{H}_2$ according to (11) or (12). Note that each inner loop with $\mathbf{H}_1$ and $\mathbf{H}_2$ optimizes $\mathbf{X}$ via ADMM. Next, we describe the algorithmic details to illustrate these improvements.

Let $\mathcal{L}(\mathbf{U},\mathbf{V},\mathbf{D}) \equiv g(\mathbf{U},\mathbf{V}) + (\mu/2)\|\mathbf{GU}+\mathbf{BV}-\mathbf{D}\|_F^2$ be the augmented Lagrangian for

$$\min_{\mathbf{U},\mathbf{V}} g(\mathbf{V}), \quad \text{s.t. } \mathbf{GU}+\mathbf{BV}=\mathbf{0} \tag{14}$$

where $\mu > 0$ is a positive constant, $\mathbf{U} = \mathbf{X}$, and $\mathbf{D} = (\mathbf{D_1},\mathbf{D_2},\mathbf{D_3},\mathbf{D_4})$ denote the Lagrange multipliers associated with the constraint $\mathbf{GU}+\mathbf{BV}=\mathbf{0}$; $\mathbf{V} \equiv (\mathbf{V_1},\mathbf{V_2},\mathbf{V_3},\mathbf{V_4})$, $\mathbf{G} = [\mathbf{A},\mathbf{I},\mathbf{I},\mathbf{I}]^{\mathbf{T}}$, $\mathbf{B} = \mathrm{diag}(-\mathbf{I})$; $g(\mathbf{V}) \equiv (1/2)\|\mathbf{V}_1-\mathbf{Y}\|_F^2 + \lambda\|(\mathbf{H}_1\mathbf{H}_2)\odot\mathbf{V}_2\|_{1,1} + \tau\sum_{s=1}^{p}\mathbf{Tr}(\mathbf{V}_{3s}\mathbf{L}_s\mathbf{V}_{3s}^T) + \iota_{R+}(\mathbf{V}_4)$ is the cost function of the following optimization problem:

$$\min_{\mathbf{U},\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3,\mathbf{V}_4} \frac{1}{2}\|\mathbf{V}_1-\mathbf{Y}\|_F^2 + \lambda\|(\mathbf{H}_1\mathbf{H}_2)\odot\mathbf{V}_2\|_{1,1} + \tau\sum_{s=1}^{p}\mathbf{Tr}\left(\mathbf{V}_{3s}\mathbf{L}_s\mathbf{V}_{3s}^T\right) + \iota_{R+}(\mathbf{V}_4)$$
$$\text{s.t. } \mathbf{V}_1=\mathbf{AU},\ \mathbf{V}_2=\mathbf{U},\ \mathbf{V}_3=\mathbf{U},\ \mathbf{V}_4=\mathbf{U} \tag{15}$$

where $\iota_{R+}(\mathbf{X}) = \sum_{i=1}^{n}\iota_{R+}(\mathbf{x}_i)$ is the indicator function and $\iota_{R+}(\mathbf{x}_i)$ is zero if $\mathbf{x}_i$ belongs to the nonnegative orthant and $+\infty$ otherwise. As shown in Algorithm 1, we execute ADMM to solve the optimization problem related to the SGSU model, the relevant Lagrange multipliers are updated sequentially in the inner loop, and the two weights are updated in the outer loop (i.e., steps 5 and 6 of Algorithm 1). In Algorithm 1, u and d_2 represent the elements in $\mathbf{U}$ and $\mathbf{D_2}$, respectively, $\mathrm{soft}(\cdot,\tau)$ represents the componentwise application of the soft-threshold function $y \mapsto \mathrm{sign}(y)\max\{|y|-\tau,0\}$.

Algorithm 1 Flowchart of the SGSU Algorithm

1: **Initialization:**
2: set $k,t = 0$, choose $\mu,\lambda,\tau,\varepsilon > 0$, $\mathbf{U}^{(0)},\mathbf{V}_1^{(0)},\mathbf{V}_2^{(0)},\mathbf{V}_3^{(0)},\mathbf{V}_4^{(0)},\mathbf{D}_1^{(0)},\mathbf{D}_2^{(0)},\mathbf{D}_3^{(0)},\mathbf{D}_4^{(0)}$, SLIC parameters
3: $\mathbf{L} = \mathbf{D}-\mathbf{W}$
4: **Repeat:**
5: $\mathbf{H}_1^{(t)} \leftarrow \mathrm{diag}\left[\frac{1}{||(\mathbf{U}^{(t)}-\mathbf{D}_2^{(t)})(1,:)||_2+\varepsilon},\ldots,\frac{1}{||(\mathbf{U}^{(t)}-\mathbf{D}_2^{(t)})(m,:)||_2+\varepsilon}\right]$
6: $\mathbf{H}_2^{(t)} \leftarrow \begin{bmatrix} h_{2,11}^{(t)} & \cdots & h_{2,1n}^{(t)} \\ \vdots & h_{2,ij}^{(t)} & \vdots \\ h_{2,m1}^{(t)} & \cdots & h_{2,mn}^{(t)} \end{bmatrix}$,
7: where $h_{2,ij}^{(t)} = \frac{1}{f_{b\in\mathcal{N}(j)}(u_{ib}^{(t)}-d_{2ib}^{(t)})+\varepsilon}$
8: **Repeat:**
9: $\mathbf{U}^{(k+1)} \leftarrow (\mathbf{A}^T\mathbf{A}+3\mathbf{I})^{-1}(\mathbf{A}^T(\mathbf{V}_1^{(k)}+\mathbf{D}_1^{(k)})+\mathbf{V}_2^{(k)}+\mathbf{D}_2^{(k)}+\mathbf{V}_3^{(k)}+\mathbf{D}_3^{(k)}+\mathbf{V}_4^{(k)}+\mathbf{D}_4^{(k)})$
10: $\mathbf{V}_1^{(k+1)} \leftarrow \frac{1}{1+\mu}[\mathbf{Y}+\mu(\mathbf{AU}^{(k+1)}-\mathbf{D}_1^{(k)})]$
11: $\mathbf{V}_2^{(k+1)} \leftarrow \mathrm{soft}(\mathbf{U}^{(k+1)}-\mathbf{D}_2^{(k)},\frac{\lambda}{\mu}\mathbf{H}_1^{(t)}\mathbf{H}_2^{(t)})$
12: for $s=1$ to p
13: $\mathbf{V}_{3s}^{(k+1)} \leftarrow \mu(\mathbf{U}_s^{(k+1)}-\mathbf{D}_{3s}^{(k)})(2\tau\mathbf{L}_s+\mu\mathbf{I})^{-1}$
14: end for
15: $\mathbf{V}_4^{(k+1)} \leftarrow \max(\mathbf{U}^{(k+1)}-\mathbf{D}_4^{(k)},0)$
16: **Update Lagrange multipliers:**
17: $\mathbf{D}_1^{(k+1)} \leftarrow \mathbf{D}_1^{(k)}-\mathbf{AU}^{(k+1)}+\mathbf{V}_1^{(k+1)}$
18: $\mathbf{D}_2^{(k+1)} \leftarrow \mathbf{D}_2^{(k)}-\mathbf{U}^{(k+1)}+\mathbf{V}_2^{(k+1)}$
19: $\mathbf{D}_3^{(k+1)} \leftarrow \mathbf{D}_3^{(k)}-\mathbf{U}^{(k+1)}+\mathbf{V}_3^{(k+1)}$
20: $\mathbf{D}_4^{(k+1)} \leftarrow \mathbf{D}_4^{(k)}-\mathbf{U}^{(k+1)}+\mathbf{V}_4^{(k+1)}$
21: **Update iteration:** $k \leftarrow k+1$
22: $\mathbf{U}^{(t+1)} \leftarrow \mathbf{U}^{(k+1)}$
23: $\mathbf{D}_2^{(t+1)} \leftarrow \mathbf{D}_2^{(k+1)}$
24: **Update iteration:** $t \leftarrow t+1$
25: **until** termination condition is satisfied.

As mentioned earlier, it is difficult to directly prove the convergence of Algorithm 1. Nevertheless, Fig. 1 shows the residual $||\mathbf{GU}^{(t)}-\mathbf{BV}^{(t)}||_F$ for Algorithm 1 decreases as the number of iterations increases. After 50 iterations, the residuals are close to 0. If a suitable maximum iteration is chosen and the algorithm's termination criteria are set such that the maximum iteration is reached or the residual is less than a certain threshold, convergence of Algorithm 1 can be guaranteed. In the following experiments, we empirically set

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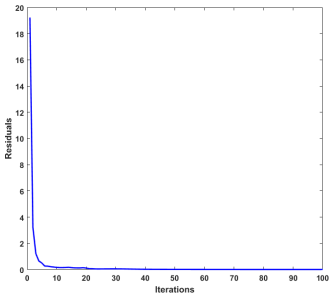


Fig. 1. Residual of Algorithm 1.

the number of inner and outer loops in the algorithm to 5 and 100, respectively. This fully guarantees the convergence of the algorithm and reduces the computational load.

IV. Experiments With Synthetic Data

In this section, we evaluate the unmixing performance of the proposed SGSU algorithm on simulated hyperspectral datasets. Two metrics are used for the quantitative analysis of unmixing results. On the one hand, it is the signal-to-reconstruction error (SRE, measured in dB) that allows us to assess the accuracy of the result, and on the other hand, the probability of success (p_s). It represents the probability estimate that the relative error power of the unmixed result is below a certain threshold. SRE (dB) is calculated as

$$\mathrm{SRE(dB)} = 10 \cdot \log_{10}\left(E\left(||\mathbf{x}||_2^2\right)/E\left(||\mathbf{x}-\widehat{\mathbf{x}}||_2^2\right)\right) \quad (16)$$

where $E(\cdot)$ denotes the expectation function, $\widehat{\mathbf{x}}$ is the estimated abundance, and $\mathbf{x}$ is the true abundance. p_s is defined as follows: $p_s \equiv P(\|\widehat{\mathbf{x}}-\mathbf{x}\|^2/\|\mathbf{x}\|^2 \leq \text{threshold})$. For instance, if we define the threshold as 10 and achieve a reconstruction accuracy p_s of 1, this means that the total relative error power of the component abundances (with a probability of 1) is below 1/10. In [5], an explanation is provided about how to set the threshold: when $\|\widehat{\mathbf{x}}-\mathbf{x}\|^2/\|\mathbf{x}\|^2 \leq 3.16$ (5 dB), the abundance estimation is considered successful. This metric can assess the stability of the abundance estimation outputs, unlike SRE (dB), which reflects the average of the estimated abundances. The larger the SRE (dB) or p_s, the better the unmixing performance of the algorithm. To illustrate the performance improvement of the proposed SGSU algorithm, the unmixing results estimated by S^2WSU [17], SUSRLR-TV [28], DRSGHU [19], and SBGLSU [20] algorithms and the parameters involved are also reported. For a fair comparison, the parameters of each algorithm were tuned before experiments.

A. Simulated Datasets

In the simulation data experiments, we used two U.S. Geological Survey (USGS) spectral libraries as mineral dictionaries. The first library $\mathbf{A_1}$ contains $m = 240$ materials (different mineral species) with reflectance in the $L = 224$ band and uniformly distributed wavelengths. It contains spectral signatures in the range from 0.4 to 2.5 μm. The second library $\mathbf{A_2}$ contains $m = 222$ spectral signatures of substances (various mineral species). Two simulated datasets are created based on these two spectral libraries, and the fractional abundances are constrained by ANC and ASC.

1) *Synthetic Data Cube 1 (DC1):* It consists of 75×75 pixels with 224 bands. Five spectral signatures were chosen from $\mathbf{A_1}$ as the end-members and the abundance map related to each end-member are generated following the method of [14]. The spectral angle distance (SAD) between any different columns (materials) in $\mathbf{A_1}$ was computed, and five spectral signatures with the minimum SAD to other columns from $\mathbf{A_1}$ were selected as the end-members for DC1. The five end-members are Jarosite GDS99 K, Sy 200C, Jarosite GDS101 Na, Sy 200, Anorthite HS349.3B, Calcite WS272, and Alunite GDS83 Na63. The scenes have been established using fractal technology, which imitates the unique spatial patterns commonly found in nature. For a visual illustration, the spectral curves of the five end-members and the true abundance maps of these end-members are displayed in Fig. 2. Then, white noise with three levels of signal-to-noise ratios (SNRs) of 30, 40, and 50 dB is added to DC1.
2) *Synthetic Data Cube 2 (DC2):* It consists of 100×100 pixels with 221 bands. Nine spectral signatures were randomly chosen from $\mathbf{A_2}$ as the end-members. The nine end-members are Alunite GDS83 Na63, Dumortierite HS190.3B, Halloysite NMNH106236, Kaolinite CM9, Kaolinite KGa-1 (wxyl), Muscovite GDS108, Nontronite GDS41, Pyrophyllite PYS1A fine g, and Sphene HS189.3B. Synthetic hyperspectral scenes with 100×100 pixels are produced following the work in [29] and [30]. The spectral curves of the nine end-members and the true abundance maps of these end-members are displayed in Fig. 3. Afterward, the simulated data cube is contaminated by i.i.d. Gaussian noise with SNRs of 30, 40, and 50 dB, respectively.

It is well known that a smaller SAD indicates higher similarity between two spectral signatures. Similar end-members tend to result in end-member mismatches and potential errors in identification. For DC1, the SAD between Jarosite GDS99 K, Sy 200C (end-member 1), and Jarosite GDS101 Na, Sy 200 (end-member 2) is 4.4445, which is the minimum SAD between all spectral signatures in $\mathbf{A_1}$. The SAD between Anorthite HS349.3B (end-member 3) and Calcite WS272 (end-member 4) is 4.4246, which is the second smallest SAD between spectral signatures in $\mathbf{A_1}$. The third smallest SAD is 4.4752, which is between Alunite GDS83 Na63 (end-member 5) and another spectral signature in the spectral library. The mean SAD between the five selected end-members is 12.6977. The mean SAD between each selected end-member and the other spectral signatures in the spectral library are 19.9283, 20.4861, 18.6093, 18.3716, and 20.4480, respectively.

For DC2, the minimum SAD between each selected end-member and other spectral signatures in the spectral library are 4.4872, 5.6978, 5.6055, 2.8740, 2.2470, 2.8075, 3.5579, 3.6531, and 5.7040. The mean SAD between the nine selected end-members is 14.8840. The mean SAD between each selected end-member and the other spectral signatures in the spectral library are 19.4677, 18.3467, 23.2971, 19.6407, 17.9256, 17.0524, 24.7773, 17.6455, and 21.1744,

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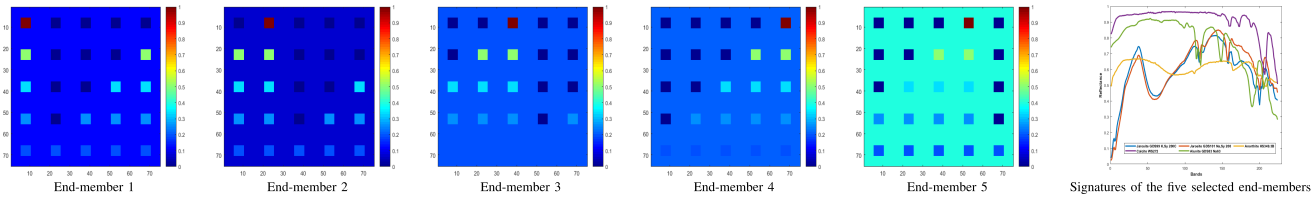


Fig. 2. Five end-member signatures and their true fractional abundances in simulated data cube 1 (DC1).

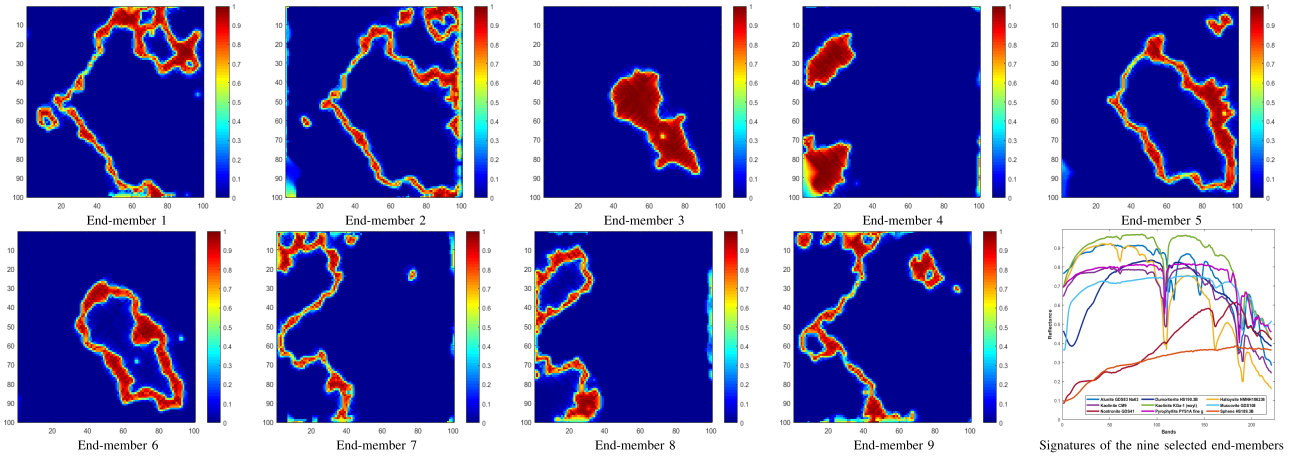


Fig. 3. Nine end-member signatures and their true fractional abundances in simulated data cube 2 (DC2).

respectively. The SAD values demonstrate that any selected end-member closely resembles a certain spectral signature in the spectral library, and these selected end-members exhibit high mutual coherence, which poses challenges in accurately identifying the end-members from the spectral library.

B. Results and Discussion

Tables I and II show the SRE (dB) and p_s values of estimated results. The best separation results obtained within the range of parameters considered are shown in the two tables, while the optimal parameters to achieve the given values are given in parentheses. When the SNR is 30, 40, and 50 dB, for the SBGLSU and SGSU algorithms, the superpixel size for the SLIC algorithm is set to 50, 50, and 25, respectively, and for the SUSRLR-TV algorithm, the size of the superpixel is set to 8, 8, and 5.

From Tables I and II, we can see that the proposed SGSU algorithm gives the best SRE (dB) and p_s results in all cases. It is worth noting that the five algorithms compared are the current cutting-edge unmixing algorithms. In the experiment with DC1, the p_s values achieved by all algorithms under all SNRs are 1, demonstrating that all algorithms can produce stable abundance estimation outputs. However, in the experiment with DC2, under the SNR of 30 dB, the p_s values attained by S^2WSU, SUSRLR-TV, and DRSGHU are below 1, slightly underperforming the results obtained by SGSU and SBGLSU. Since there are many small subabundances in DC1, it is difficult to accurately represent the correlation between abundance vectors of different pixels by relying only on neighborhood weighting strategies, and the result of S^2WSU is clearly bad. The graph manifold structure reflects pixel nuances to obtain more accurate abundance solutions. Compared with the DRSGHU algorithm, the SBGLSU and SGSU algorithms achieve higher SRE (dB) values with lower SNR due to the superpixel segmentation of hyperspectral images. It has been proven that flexible spatial clustering strategies can make better use of image spatial information and improve the noise robustness of algorithms. The true abundance of DC2 is partially smooth and fairly sparse. With the introduction of spatial neighborhood weights, the S^2WSU and SGSU algorithms clearly outperform the other algorithms. Compared with the S^2WSU algorithm, the proposed SGSU algorithm exhibits better unmixing accuracy, suggesting that spatial weights guided by local spectral similarity are more effective.

Due to the presence of other spectral signatures in the spectral library that is highly similar to the selected end-members, the unmixing algorithm is prone to incorrectly identifying these similar spectral signatures as the correct end-members. To further illustrate the advantages of the proposed SGSU algorithm in end-member identification, Fig. 4 shows the abundance maps obtained by applying different unmixing algorithms to DC2. The rows with nonzero values in abundance maps indicate the corresponding end-members in the spectral library that generate the hyperspectral images. It should be noted that 500 pixels were randomly selected from all pixels to display the corresponding abundance map to present the results more clearly. As can be seen from Fig. 4, all algorithms found the correct end-members from the spectral library, but results produced by SGSU are the closest to the true abundances. Compared with other algorithms, the abundance map obtained by the SGSU algorithm is the cleanest in the area where the abundance value should be zero, indicating that the SGSU algorithm obtains the least erroneous end-members. The DRSGHU algorithm obviously found three erroneous end-members. The abundance maps of SUSRLR-TV and SBGLSU algorithms also indicate erroneous

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TABLE I

SRE (dB) AND p_s VALUES OF THE RESULTS OBTAINED BY DIFFERENT ALGORITHMS ON DC1 (THE OPTIMAL PARAMETERS FOR THE REPORTED RESULTS ARE LISTED IN PARENTHESES)

Algorithm	SNR=30dB		SNR=40dB		SNR=50dB	
	SRE(dB)	p_s	SRE(dB)	p_s	SRE(dB)	p_s
S^{2}WSU	16.6657	1	26.5792	1	36.5915	1
	($\lambda = 2e$-3)		($\lambda = 3e$-4)		($\lambda = 1e$-4)	
SUSRLR-TV	25.2914	1	42.8581	1	51.3169	1
	($\rho = 1e$-1,$\lambda_{TV} = 1e$-2)		($\rho = 1e$-2,$\lambda_{TV} = 5e$-4)		($\rho = 5e$-3,$\lambda_{TV} = 1e$-4)	
DRSGHU	29.5571	1	52.8699	1	63.1721	1
	($\alpha_1 = 1e3$,$\alpha_2 = 9e$-4)		($\alpha_1 = 1e3$,$\alpha_2 = 1e$-3)		($\alpha_1 = 1e3$,$\alpha_2 = 7e$-4)	
SBGLSU	33.3644	1	53.1373	1	61.8580	1
	($\lambda_s = 6e$-2,$\lambda_g = 3e2$)		($\lambda_s = 4e$-3,$\lambda_g = 1e3$)		($\lambda_s = 4e$-3,$\lambda_g = 1e3$)	
SGSU	**34.9117**	1	**54.6418**	1	**64.1684**	1
	($\lambda = 1e$-3,$\tau = 2e2$)		($\lambda = 5e$-5,$\tau = 1e3$)		($\lambda = 6e$-5,$\tau = 1e3$)	

TABLE II

SRE (dB) AND p_s VALUES OF THE RESULTS OBTAINED BY DIFFERENT ALGORITHMS ON DC2 (THE OPTIMAL PARAMETERS FOR THE REPORTED RESULTS ARE LISTED IN PARENTHESES)

Algorithm	SNR=30dB		SNR=40dB		SNR=50dB	
	SRE(dB)	p_s	SRE(dB)	p_s	SRE(dB)	p_s
S^{2}WSU	26.9556	0.9995	37.7721	1	45.5675	1
	($\lambda = 5e$-3)		($\lambda = 2e$-3)		($\lambda = 4e$-4)	
SUSRLR-TV	19.9805	0.9980	26.1654	1	29.4822	1
	($\rho = 3e$-3,$\lambda_{TV} = 2e$-2)		($\rho = 1e$-3,$\lambda_{TV} = 9e$-4)		($\rho = 5e$-4,$\lambda_{TV} = 4e$-4)	
DRSGHU	8.7091	0.8695	28.1855	1	43.0854	1
	($\alpha_1 = 1e1$,$\alpha_2 = 1e$-2)		($\alpha_1 = 8e$-3,$\alpha_2 = 2e$-3)		($\alpha_1 = 5e$-3,$\alpha_2 = 2e$-3)	
SBGLSU	24.8025	1	34.9372	1	42.0948	1
	($\lambda_s = 4e$-2,$\lambda_g = 9e$-1)		($\lambda_s = 6e$-2,$\lambda_g = 2e$-1)		($\lambda_s = 3e$-2,$\lambda_g = 3e$-3)	
SGSU	**29.7282**	1	**39.3263**	1	**46.4574**	1
	($\lambda = 3e$-3,$\tau = 5e$-1)		($\lambda = 2e$-3,$\tau = 8e$-2)		($\lambda = 4e$-4,$\tau = 1e$-3)	

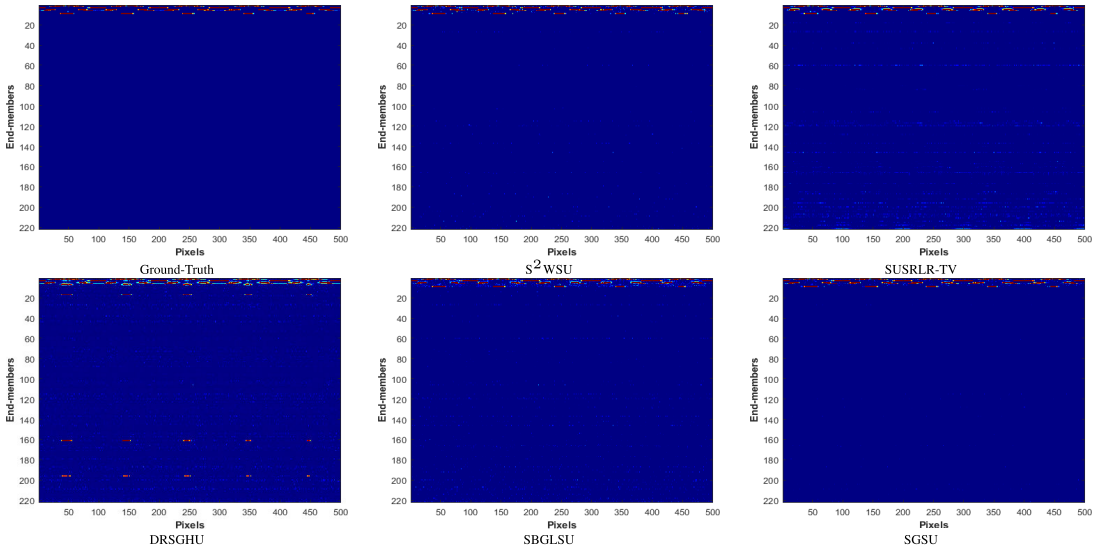


Fig. 4. True abundances and estimated abundances obtained by different unmixing algorithms for 500 pixels abundances in simulated data cube 2 (DC2) at SNR = 30 dB.

end-members to different extents, although the abundance values of erroneous end-members are smaller than those obtained by the DRSGHU algorithm. The S^{2}WSU algorithm obtained some small outliers where the true abundance is zero, but the results are better than other comparison methods, which proves the effectiveness of the spatial–spectral weighting strategy.

Moreover, the selected end-members with high mutual coherence are prone to end-member mismatches. For intuitive comparison, Figs. 5 and 6 show abundance estimates of the resulting unmixing algorithm taken into account, performed at DC1, simulated at SNR = 30 dB. The graphical comparison shows only the abundance maps of the first end-member in DC1 and the eighth end-member in DC2, as the abundance maps behave similarly. These figures also show the difference between estimated and true abundances. The abundance maps obtained by the SBGLSU and SGSU algorithms are close to the true abundance, especially for the SGSU algorithm,

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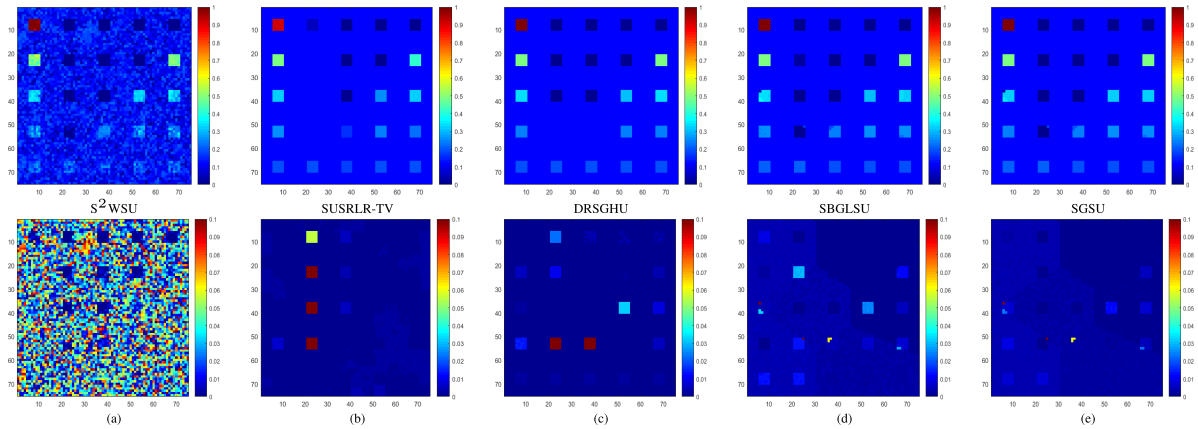


Fig. 5. Abundance maps obtained for the first end-member from DC1 under the noise ratio of SNR = 30 dB and difference maps between the ground-truth abundances and the result obtained by compared algorithms at SNR = 30 dB. (a) S^2WSU. (b) SUSRLR-TV. (c) DRSGHU. (d) SBGLSU. (e) SGSU.

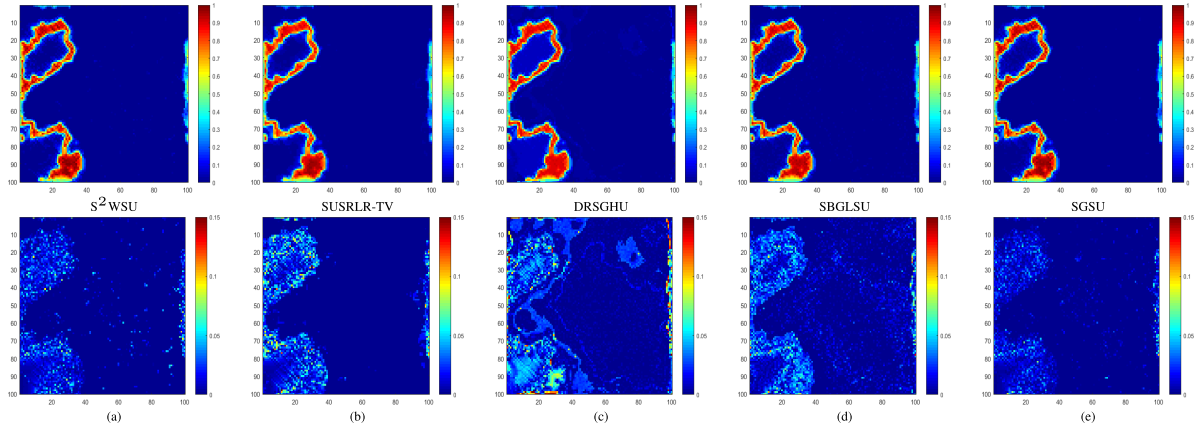


Fig. 6. Abundance maps obtained for the eighth end-member from DC2 under the noise ratio of SNR = 30 dB and difference maps between the ground-truth abundances and the result obtained by compared algorithms at SNR = 30 dB. (a) S^2WSU. (b) SUSRLR-TV. (c) DRSGHU. (d) SBGLSU. (e) SGSU.

which can be seen from the difference map closest to the true abundance. However, from the difference maps, we find that the abundance maps obtained by the proposed method are more detailed than those produced by the SBGLSU algorithm, preserving fine structure and texture, and producing better abundance estimates. The SGSU algorithm concentrated the nonzero abundances on the selected end-members, achieving more accurate abundance distribution estimation. The abundance map obtained by the S^2WSU algorithm is considered noisy and inaccurate, especially at DC1. The DRSGHU algorithm is not accurate enough to identify end-member distributions and incorrect end-members are shown in the abundance maps. This observation highlights that the integration of local similarity spatial weighting strategy and superpixel-based spatial regularization utilizes multiscale spatial information to improve the ability of unmixing algorithms to correctly identify end-members and maintain spatial correlation on the abundance map. This further shows that spatial weighting of local spectral similarities can improve unmixing performance.

C. Analysis of the Computational Cost

We analyze the computational complexity of the algorithms involved in this experiment. Table III shows the processing time (in seconds) for different algorithms. All unmixing algorithms are performed in MATLAB R2016a using a desktop computer with an Intel Core i7-6700 3.4 GHz CPU and 24 GB memory bank. As shown in Table III, the graph regularization method takes longer than the S^2WSU method. However, compared to the DRSGHU method, the SBGLSU and SGSU methods implement graph regularization after image segmentation, which effectively reduces the computational cost.

D. Ablation Study

The proposed SGSU model includes sparse regularization with locally similar guidance and graph regularization based on superpixels. To analyze the effects of these two regularizers more clearly, we conducted further experiments. First, we compared the influence of superpixel segmentation in the graph regularization term on the accuracy of abundance estimation. As a special case of the SGSU algorithm, we applied global graph regularization to the entire abundance image without superpixel segmentation, which we refer to as the SGSU-1 algorithm. In addition, to demonstrate the effect of the new local spatial similarity-guided sparse regularization, we present another special case of the SGSU algorithm, which removes the graph regularization term and only retains the

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TABLE III
RUN TIMES (IN SECONDS) OF DIFFERENT UNMIXING METHODS ON PROCESSING DC1 WHEN SNR = 30 DB

Algorithm	S^2WSU	SUSRLR-TV	DRSGHU	SBGLSU	SGSU
Time(s)	93.41	377.09	233.88	181.38	191.70

TABLE IV
SRE (DB) VALUES AND RUN TIMES (IN SECONDS) OBTAINED BY RUNNING SGSU, SGSU-1, AND SGSU-2 ALGORITHMS ON DC2 (THE OPTIMAL PARAMETERS FOR THE REPORTED RESULTS ARE LISTED IN PARENTHESES)

SNR		30dB	40dB	50dB
SGSU-1	SRE(dB)	29.9758	39.5715	46.6005
		($\lambda = 3e$-3,$\tau = 5e$-1)	($\lambda = 2e$-3,$\tau = 8e$-2)	($\lambda = 4e$-4,$\tau = 1e$-3)
	Time(s)	263.42	271.62	270.15
SGSU-2	SRE(dB)	27.4371	38.2587	46.1067
		($\lambda = 5e$-3)	($\lambda = 2e$-3)	($\lambda = 4e$-4)
	Time(s)	77.74	77.67	78.04
SGSU	SRE(dB)	29.7282	39.3263	46.4574
		($\lambda = 3e$-3,$\tau = 5e$-1)	($\lambda = 2e$-3,$\tau = 8e$-2)	($\lambda = 4e$-4,$\tau = 1e$-3)
	Time(s)	191.70	193.01	193.95

spatially weighted sparse constraint. This algorithm is referred to as the SGSU-2 algorithm. Table IV shows the SRE (dB) values and run times for the abundance estimation obtained by running SGSU-1, SGSU-2, and SGSU algorithms on DC2. As can be seen from Table IV, the SGSU-1 algorithm outperforms the SGSU algorithm by a slight margin and achieves the best results. However, the runtime of the SGSU-1 algorithm is approximately 1.4 times longer than that of the SGSU algorithm. Therefore, it can be inferred that choosing the appropriate size for superpixels can approximate the optimal unmixing accuracy and significantly reduce the running time. The SGSU-2 algorithm performs similarly to the other two algorithms under high SNR, but relatively poorly under low SNR, indicating that graph regularization has some effectiveness in improving the algorithm's noise resistance. Overall, for the DC2 dataset, algorithms that employ spatial weighting strategies, such as the S^2WSU and SGSU series algorithms, perform better than those that do not use spatial weighting. Furthermore, by comparing the estimated abundance of the SGSU-2 and S^2WSU algorithms, it is known that spatial weighting guided by local similarity provides a more accurate characterization of spatial correlations between pixels than spatial weighting using Euclidean distance, which leads to better performance.

V. REAL DATA EXPERIMENTS

Within this division, we employ actual Cuprite data and Mangrove data to verify the effectiveness of the proposed SGSU algorithm.

A. Cuprite Data

In this section, we evaluate the proposed method using a real Cuprite dataset acquired by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS). This dataset is a common benchmark for validating hyperspectral separation algorithms and is available online at.[1] For this experiment, we select a portion of this scene that is 350 × 350 pixels in size. This dataset originally had 224 bands ranging from 0.4 to 2.5 μm, and the spectral resolution reached 10 nm. Before conducting experiments, bands 1–2, 105–115, 150–170, and 223–224 were excluded due to the low SNR and water absorption effects, and only 188 bands were retained. In this experiment, we use the same spectral library $\mathbf{A_1}$ as DC1, but we filter out the defective bands as before, leaving only the same 188 spectral bands as the dataset.

Fig. 7 displays the different mineral classification maps for the Cuprite mining area. Drawn by the Tricorder software,[2] this mineral map was produced by the USGS in 1995 using the Tricorder 3.3 software product [31], which usually serves as a reference for the result of unmixing the Cuprite dataset to qualitatively evaluate the estimated abundance maps obtained by different unmixing algorithms. The true abundances of the minerals in the Cuprite dataset are not available. Therefore, we can only qualitatively evaluate the performance of different unmixing algorithms via the contrast of the estimated abundance and the reference map.

Fig. 8 qualitatively compares the classification map generated by the USGS Tricorder algorithm with the abundance maps estimated by S^2WSU, SUSRLR-TV, DRSGHU, SBGLSU, and SGSU algorithms for three typical minerals (Alunite, Buddingtonite, and Chalcedony). For the regularization parameter settings in the experiment, S^2WSU, SUSRLR-TV, DRSGHU, SBGLSU, and SGSU were empirically set to $\lambda = 0.002$, ($\rho = 0.0005, \lambda_{\text{TV}} = 0.005$), ($\alpha_1 = 0.5, \alpha_2 = 0.2$), ($\lambda_s = 0.1, \lambda_g = 0.05$), and ($\lambda = 0.001, \tau = 0.5$), respectively. As shown in Fig. 8, all the unmixing algorithms obtained reasonable abundance estimation results.

[1] http://aviris.jpl.nasa.gov/html/aviris.freedata.html
[2] http://speclab.cr.usgs.gov/PAPER/tetracorder

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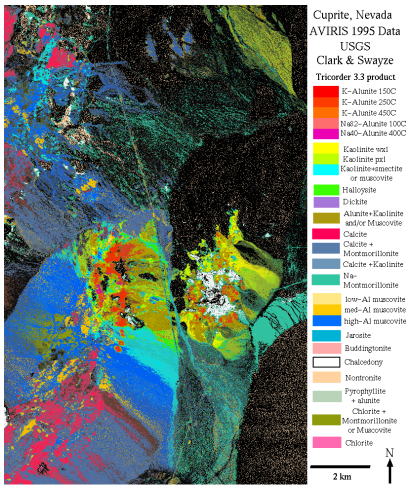


Fig. 7. USGS map showing the location of different minerals in the Cuprite mining district in Nevada.

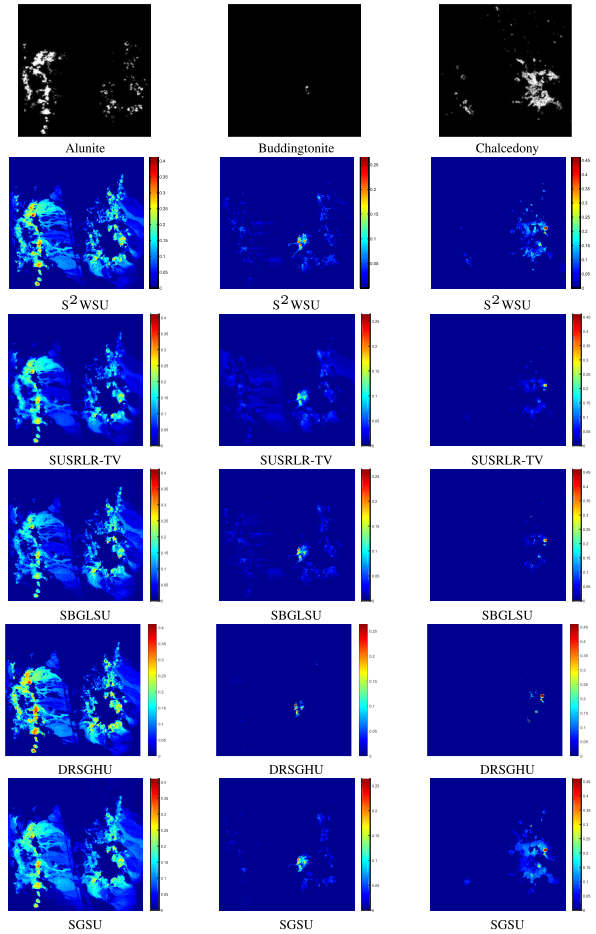


Fig. 8. Fractional abundance maps estimated by S^2WSU, SUSRLR-TV, SBGLSU, DRSGHU, and SGSU and the classification maps produced by USGS Tricorder software for the considered 350 × 350-pixel subset of the AVIRIS Cuprite scene.

The abundance maps explicitly show the presence of the considered minerals, confirming that these sparse unmixing methods are effective in interpreting the real datasets considered. However, some abundance maps appear noisy, such as those obtained from the S^2WSU, DRSGHU, and SBGLSU algorithms for the Buddingtonite mineral, but those obtained from SUSRLR-TV are oversmoothed and the SGSU estimated abundance map is the closest to the reference map. In addition, the SGSU algorithm provides abundance maps with improved spatial consistency for major minerals (such as buddingtonite and chalcedony). Overall, the abundance maps estimated by the SGSU algorithm are generally comparable to or better than those produced by the other algorithms. Therefore, from a qualitative point of view, we can conclude that the newly developed SGSU approach is likely to improve the unmixing results in practical scenarios.

We analyzed the computational complexity of algorithms used in the Cuprite real data experiment. Table V presents the processing time (in seconds) for different algorithms. All algorithms were run on a desktop computer equipped with an Intel Core i7-13700KF 3.4 GHz CPU and 32 GB of memory using MATLAB R2023b. According to the results in Table V, among the three graph regularization methods, the DRSGHU method required the most time to process, and this result was obtained by employing a block processing approach. It is worth noting that attempting to apply this method directly to the entire image for global processing would overload the device during the construction of the global adjacency matrix, making it impossible to handle the computation. In contrast, the SGSU and SBGLSU methods, which utilize superpixel block processing, can process the image at a faster speed, significantly reducing the computational cost. Furthermore, compared to the SUSRLR-TV method, the SGSU method exhibits higher computational efficiency.

B. Mangrove Data

The data utilized for this experiment was obtained in October 2016 using an unmanned aerial vehicle equipped with the Cubert UHD185 hyperspectral imaging system. The imaging took place in the mangrove nature reserve located on Qi'ao Island, Zhuhai City, China, as depicted in Fig. 9(a) [32], [33]. The empirical assay scenario covers an area of 1878 × 1877 pixels, spanning the spectral range of 450–950 nm with spectral discrimination of 4 nm and containing 125 spectral bands. The imagistic evidential data encompasses an approximate area of 3 hectares within the mangrove biomes. After undergoing preliminary processing, the images were resampled to achieve a spatial resolution of 0.02 m.

In this study, we documented the major mangrove species in the research area, including Kandelia candel (KC), Acanthus aureum (AA), Aegiceras corniculatum (AC), Sonneratia apetala (SA), Acanthus ilicifolius (AI), Hymenocallis littoralis (HL), and Thespesia populnea (TP), based on [32] and [33]. Additionally, the area also contained other land features such as Phragmites australis (PA), tidal creeks, wooden footpaths, and a specific category for shading. It is important to note that the spectral library $\mathbf{A}_3 \in \mathbb{R}^{125\times 23}$ used in this study is consistent with the one described in [34] and [35]. Fig. 9(b) displays the spectral characteristics of the seven main vegetation species included in this library. From the figure, it is evident that the spectral signatures of different vegetation categories exhibit high similarity, thus posing considerable challenges for spectral unmixing. To qualitatively assess the

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TABLE V
RUN TIMES (IN SECONDS) OF VARIOUS UNMIXING METHODS WHEN PROCESSING THE CUPRITE DATA

Algorithm	S^2WSU	SUSRLR-TV	DRSGHU	SBGLSU	SGSU
Time(s)	163.38	2319.75	1162.48	362.39	426.12

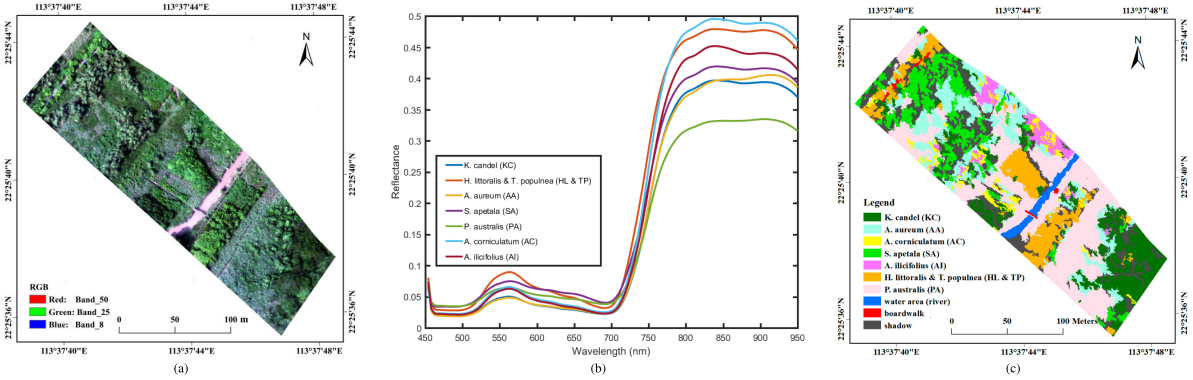


Fig. 9. Mangrove hyperspectral data. (a) UHD185 hyperspectral image. (b) Spectral reflectance profiles of seven vegetation species. (c) Classification map of UHD185 hyperspectral image.

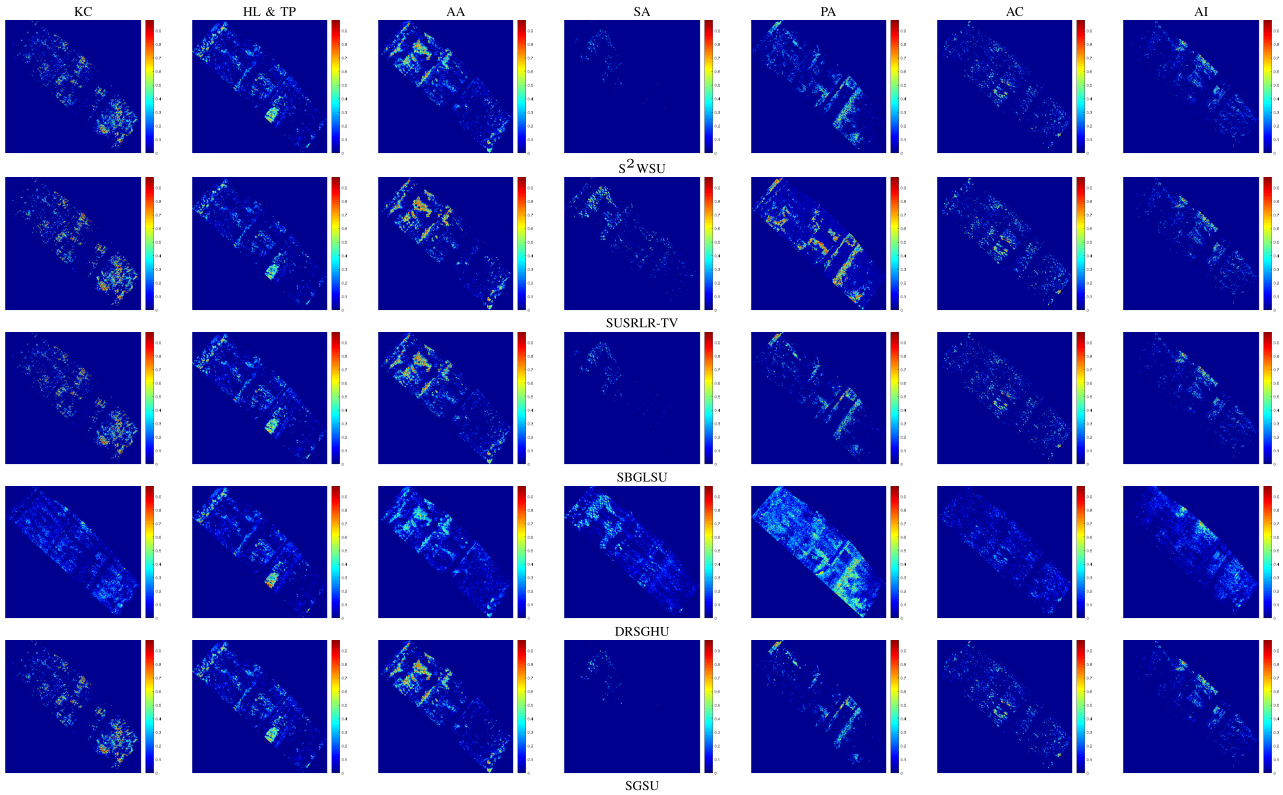


Fig. 10. Abundance maps of seven vegetation types produced by each sparse unmixing algorithm.

performance of various algorithms in unmixing this data, we referred to the land-cover map of the research area created by Cao et al. [32] using a support vector machine classification method [refer to Fig. 9(c)].

In this experiment, the regularization parameters for the S^2WSU, SUSRLR-TV, DRSGHU, SBGLSU, and SGSU algorithms were empirically set to $\lambda = 1e\text{-}4$, ($\rho = 1e\text{-}4$, $\lambda_{\text{TV}} = 1e\text{-}5$), ($\alpha_1 = 3e\text{-}5, \alpha_2 = 3e\text{-}5$), ($\lambda_s = 1e\text{-}5, \lambda_g = 1e\text{-}5$), and ($\lambda = 1e\text{-}4, \tau = 1e\text{-}4$), respectively. All algorithms were run on the same computing platform as used for processing the Cuprite dataset. It is important to emphasize that the DRSGHU method incurs a significant computational burden and the computing server struggles to perform the calculations when the block processing approach is not adopted. As a result, no results can be obtained. In contrast, the SGSU and SBGLSU methods using superpixel block processing were able to process the data at a faster speed. This further demonstrates the efficacy of our proposed method in handling

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large-scale data. Furthermore, it should be noted that the abundance results estimated by the DRSGHU method shown in Fig. 10 were based on the same superpixel segmentation results as those used in the SGSU and SBGLSU methods.

In accordance with Fig. 10, the presented results demonstrate the typical vegetation abundance maps estimated by different algorithms, encompassing seven distinct vegetation types: KC, HL & TP, AA, SA, PA, AC, and AI. The regions within the image exhibiting abundance values greater than 0 indicate the presence of the corresponding vegetation types. Observing Fig. 10, it becomes apparent that the outcomes obtained from these five sparse unmixing algorithms generally match the reference classification map in most regions. This confirms the efficacy of sparse unmixing algorithms in effectively discerning between different end-members within the dataset. Remarkably, the visually discernible effects indicate a high consistency between the abundance maps generated by the proposed SGSU algorithm and the reference classification map. The maps generated by the SGSU algorithm clearly depict the distribution of the majority of vegetation species, with higher abundance values in vegetated areas, while retaining more spatial information. By contrast, although the S^2WSU algorithm indicates the existence of PA vegetation, the proportion is not high, and the SUSRLR-TV algorithm clearly obtains irrelevant small abundance values. In fact, the DRSGHU algorithm requires the establishment of a Laplacian matrix for the entire mangrove hyperspectral image, which exceeds the matrix size allowed by MATLAB, making it impossible to execute directly. We had to adopt the same grouping strategy as the SGSU and SBGLSU algorithms, so the results of these three algorithms are somewhat similar. However, the SGSU algorithm retains more spatial information, particularly evident in delicate details such as the KC vegetation. This evidence supports the advantages of the SGSU algorithm in handling mangrove datasets.

VI. Conclusion and Future Developments

In this article, we present a novel sparse unmixing algorithm for hyperspectral images using spatial graph regularization based on local spectral similarity. The proposed SGSU flexibly uses spatial information via weighting factors and graph regularization, whereas one of the weighting factors is guided by the spectral similarity of pixels within the local window, while the hyperspectral image is divided into different superpixel blocks according to the spectral distance and the spatial distance before the graph is constructed. A related optimization problem is solved iteratively in a double-loop structure according to the ADMM framework. The outer loop recomputes the two weighting factors and the inner loop updates the separation factor. In experiments on synthetic and real hyperspectral data, the SGSU algorithm consistently shows superior unmixing performance compared to other advanced algorithms. Although this work yielded promising experimental results, we will perform further experiments with additional datasets to fully evaluate the SGSU algorithm. In future work, we will further optimize the SGSU hyperspectral unmixing model using new technologies such as deep learning [36], [37], [38] and parallel computing [39], [40].

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